

## ODD 2024 Tutorial Sheet -8

### Software Development Fundamentals – I (15B11CI111)

| Course Outcomes (CO)                                                 |                                                                                                                                             |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CO1</b> Explain various phases of software development life cycle |                                                                                                                                             |
| <b>CO2</b>                                                           | Explain various data types, memory allocation schemes, precedence of arithmetical and logical operations, and need of array, and structures |
| <b>CO3</b>                                                           | Draw the flow chart and write the high level code for different problems                                                                    |
| <b>CO4</b>                                                           | Apply and implement functions with or without pointers for different problems                                                               |
| <b>CO5</b>                                                           | Demonstrate and implement various operations like traverse, insertion, deletion, etc. on files                                              |

**Note:** Students are advised to submit their solutions to their respective tutorial faculties

Q1. [**CO4**] Use pointer to delete all occurrences of a given character in the string. The program should output the corrected string with no holes.

Q2. [**CO4**] Write a program to swap two numbers using a pointer.

Q3. [**CO4**] What will be the output of the program?

```
a) int main()
{
    int i=3, *j, k;
    j = &i;
    printf("%d\n", i**j*i+j);
    return 0;
}

b) int main()
{
    int x=30, *y, *z;
    y=&x; /* Assume address of x
    is 500 and integer is 4 byte
```

```
size */
    z=y;
    *y++=*z++;
    x++;
    printf("x=%d, y=%d, z=%d\n",
    x, y, z);
    return 0;
}

c) int main()
{
    char str[20] = "Hello";
    char *const p=str;
```

```

        *p='M';
        printf("%s\n", str);
        return 0;
    }

d) int main()
{
    int ***r, **q, *p, i=8;
    p = &i;
    q = &p;
    r = &q;
    printf("%d, %d, %d\n", *p,
        **q, ***r);
    return 0;
}

e) int main()
{
    char *str;
    str = "%s";
    printf(str, "K\n");
    return 0;
}

f) int main()
{
    printf("%d,%d\n",
        sizeof(NULL), sizeof(""));
    return 0;
}

g) int main()
{
    void *vp;
    char ch=74, *cp="JACK";
    int j=65;
    vp=&ch;
    printf("%c", *(char*)vp);
    vp=&j;
    printf("%c", *(int*)vp);
    vp=cp;
    printf("%s", (char*)vp+2);
    return 0;
}

h) int main()

```

```

{
    int arr[2][2][2] = {10, 2, 3,
        4, 5, 6, 7, 8};
    int *p, *q;
    p = &arr[1][1][1];
    q = (int*) arr;
    printf("%d, %d\n", *p, *q);
    return 0;
}

i) int main()
{
    int a[3][4] = { 1, 2, 3, 4,
        5, 6, 7, 8, 9, 10, 11, 12 };

    printf("%u,%u,%u\n", a[0]+1,
        *(a[0]+1), (*(a+0)+1));

    return 0;
}

Assume that the array begins at location 1002

j) int main()
{
    int arr[3] = {2, 3, 4};
    char *p;
    p = arr;
    p = (char*)((int*)(p));
    printf("%d, ", *p);
    p = (int*)(p+1);
    printf("%d", *p);
    return 0;
}

k) int main()
{
    char *str;
    str = "%d\n";
    str++;
    str++;
    printf(str-2, 300);
    return 0;
}

l) int main()
{
    char str[] = "peace";
    char *s = str;

```

```

        printf("%s\n", s++ +3);
        return 0;
    }

m) int main()
{
n) int main()
{
    char *p;
    p="hello";
    printf("%s\n", *&p);
    return 0;
}
}

```

Assume that the array begins at location 1002

```

o) int main()
{
    char str1[] = "India";
    char str2[] = "BIX";
    char *s1 = str1, *s2=str2;
    while(*s1++ = *s2++)
        printf("%s", str1);
    printf("\n");
    return 0;
}

```

```

p) #include<stdio.h>
#include<string.h>
int main()
{
    int i, n;
    char *x="Alice";
    n = strlen(x);
    for(i=0; i<=n; i++)
    {
        printf("%s ", x);
        x++;
    }
    printf("\n", x);
    return 0;
}

```